HapticSense — Adaptive Haptic Navigation Belt for Visually Impaired Users

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***Abstract*—This paper reviews wearable obstacle detection systems and vibrotactile belt interfaces for visually impaired navigation and formulates research questions and hypotheses for evaluating a low-cost adaptive haptic navigation belt.**

***Index Terms*—Wearable assistive technology, haptic feedback, vibrotactile belt, obstacle detection, ultrasonic sensor, visually impaired navigation**

I. Literature Review

A. Overview of Assistive Navigation Systems

Assistive navigation systems for visually impaired individuals have gained significant attention in recent years. These systems typically utilize sensors such as ultrasonic, infrared, or camera-based modules to detect obstacles and provide feedback to the user. Among these, ultrasonic sensor-based systems are widely adopted due to their low cost, simplicity, and reliability in real-time distance measurement [1], [3].

Most existing systems focus on detecting the presence of obstacles and estimating their distance from the user. However, they primarily rely on static measurements, which limits their effectiveness in dynamic environments where both the user and obstacles may be in motion.

B. Ultrasonic Sensing for Obstacle Detection

Ultrasonic sensors operate based on the time-of-flight (ToF) principle, where the distance is calculated from the time taken for an emitted acoustic signal to reflect back from an obstacle. This technique has been widely used in wearable assistive devices for obstacle detection and navigation [1].

While ultrasonic sensing provides accurate distance estimation, most implementations do not consider temporal variations in sensor readings. As a result, they lack the ability to interpret motion behavior such as approaching or receding obstacles.

C. Wearable Assistive Devices

Wearable assistive systems integrate sensing, processing, and feedback mechanisms to enhance user mobility. Previous studies have demonstrated the effectiveness of such systems in improving obstacle awareness and navigation safety [3].

However, these systems generally operate on threshold-based logic, where feedback is triggered when the detected distance falls below a predefined value. This approach does not account for dynamic risk factors, such as the speed of approach or the likelihood of collision.

D. Vibrotactile Feedback for Navigation

Haptic feedback, particularly vibrotactile feedback, has been extensively studied as an effective means of conveying spatial information. Research has shown that directional vibration patterns can guide users effectively without relying on visual or auditory cues [2], [4].

Despite its effectiveness, most existing implementations use fixed vibration patterns based solely on distance. There is limited exploration of adaptive feedback strategies that incorporate dynamic environmental factors such as motion and risk.

E. Limitations of Existing Approaches

From the literature, the following limitations are identified:

- Dependence on static distance measurements without temporal analysis
- Lack of motion interpretation (approaching or receding obstacles)
- Absence of predictive metrics such as collision risk and time-to-collision
- Limited adaptability in haptic feedback mechanisms

These limitations highlight the need for systems that can interpret environmental dynamics rather than relying solely on proximity-based detection.

F. Proposed Contribution

To address these gaps, the proposed HapticSense system introduces a dynamic environmental interpretation framework. By utilizing time-stamped distance data, the system computes relative speed, collision risk, and time-to-collision (TTC).

This approach enables the system to move beyond basic obstacle detection and toward predictive sensing. Additionally, the integration of adaptive haptic feedback based on risk and TTC enhances user awareness and safety.

The proposed methodology aligns with emerging trends in intelligent assistive navigation systems, where combining spatial and temporal analysis leads to improved performance and usability [3], [4].

II. Research Questions and Hypotheses

A. Research Motivation

Traditional assistive navigation systems primarily rely on static distance measurements for obstacle detection. However,

real-world environments are dynamic, where both the user and obstacles may be in motion. Therefore, there is a need to investigate whether incorporating temporal information such as speed, collision risk, and time-to-collision (TTC) can improve the effectiveness of assistive systems.

Existing wearable assistive devices primarily focus on proximity-based feedback, with limited consideration of dynamic environmental interpretation [1], [3]. Additionally, vibrotactile feedback has been shown to be effective in conveying spatial information, but its adaptation to dynamic risk conditions remains an open research area [2], [4].

B. Research Questions

Based on the proposed dynamic sensing framework, the following research questions are formulated:

- **RQ1:** Can ultrasonic sensor-based systems provide accurate real-time distance measurement for obstacle detection?
- **RQ2:** Does incorporating temporal analysis (relative speed) improve the system's ability to interpret dynamic environmental conditions?
- **RQ3:** Can collision risk and time-to-collision (TTC) be effectively estimated using distance and speed data?
- **RQ4:** Does adaptive haptic feedback based on risk and TTC improve user awareness compared to static distance-based feedback?
- **RQ5:** How accurately can the system classify obstacle direction (Left, Right, Both, None) under varying conditions?

C. Hypotheses

To address the research questions, the following hypotheses are proposed:

- **H1:** The ultrasonic sensor system can measure obstacle distance with acceptable accuracy for real-time applications, consistent with prior assistive sensing systems [1].
- **H2:** Relative speed derived from time-series distance data enables identification of dynamic obstacle behavior (approaching or receding), extending traditional static detection methods [3].
- **H3:** Collision risk and TTC metrics provide meaningful indicators of potential hazards in dynamic environments, similar to predictive safety approaches in navigation systems [2].
- **H4:** Adaptive haptic feedback based on risk and TTC results in more effective and intuitive user alerts compared to fixed-threshold feedback, aligning with vibrotactile guidance principles [4].
- **H5:** The system achieves reliable directional classification with high accuracy, as supported by confusion matrix and performance metrics.

D. Expected Outcomes

It is expected that the integration of temporal and spatial analysis will enhance the system's capability beyond basic obstacle detection. The system should not only detect obstacles but also interpret motion behavior, predict potential collisions, and provide adaptive feedback.

Such an approach aligns with emerging trends in intelligent assistive navigation systems, where predictive and context-aware feedback significantly improves user safety and situational awareness [3], [4].

III. Design and Implementation

A. System Overview

The proposed HapticSense system is designed as a real-time assistive device that combines ultrasonic sensing, embedded processing, and haptic feedback to provide directional awareness. Unlike conventional systems that rely solely on instantaneous distance measurement, this design integrates temporal analysis to extract dynamic environmental information such as relative speed, collision risk, and time-to-collision (TTC).

The system consists of three major modules: sensing, processing, and feedback, following established wearable assistive system architectures [1], [3].

B. Hardware Architecture

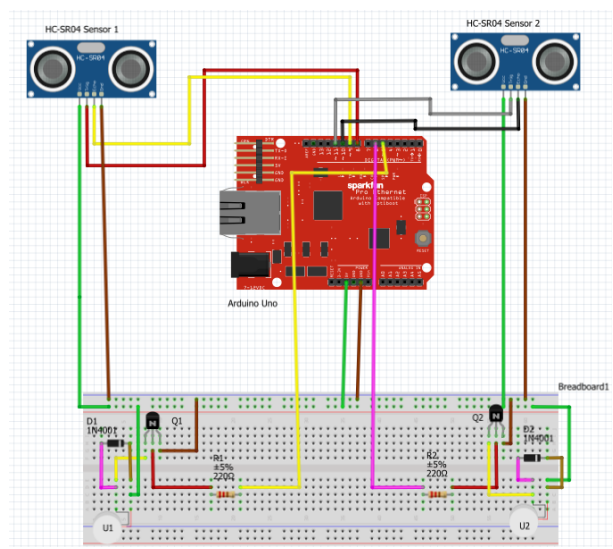

Fig. 1. Circuit diagram of the HapticSense system showing dual ultrasonic sensors, Arduino microcontroller, and vibration motor driver circuits

The hardware implementation consists of two HC-SR04 ultrasonic sensors, an Arduino Uno microcontroller, and two vibration motors driven through transistor-based switching circuits.

Sensing Unit: Two ultrasonic sensors are placed on the left and right sides to detect obstacles. Each sensor operates using the time-of-flight (ToF) principle, where the time taken for an emitted ultrasonic pulse to return is measured and converted into distance, a method widely used in assistive navigation systems [1].

Processing Unit: An Arduino Uno microcontroller processes sensor inputs, computes distance, and performs real-time data analysis including speed estimation, collision risk calculation, and TTC evaluation.

Feedback Unit: Two vibration motors provide directional haptic feedback. The motors are controlled using NPN transistors (Q1 and Q2) to allow sufficient current drive, with protection diodes (D1, D2) to prevent back EMF damage. Vibrotactile feedback has been shown to be effective for spatial guidance in assistive devices [2], [4].

C. Working Principle

The system measures distance using ultrasonic time-of-flight:

$$d = \frac{v \cdot t}{2} \tag{1}$$

where d is distance, v is the speed of sound, and t is the measured echo time.

To capture dynamic behavior, consecutive measurements are used to compute relative speed:

$$v_r = \frac{d_{i-1} - d_i}{\Delta t} \tag{2}$$

This enables detection of approaching or receding obstacles, extending traditional static sensing approaches [3].

D. Dynamic Environmental Interpretation

The system extends beyond basic sensing by incorporating advanced metrics:

Collision Risk:

$$R_c = \frac{|v_r|}{d} \tag{3}$$

Time-to-Collision (TTC):

$$TTC = \frac{d}{|v_r|} \tag{4}$$

These parameters allow the system to predict hazardous situations rather than react only to proximity, similar to techniques used in intelligent navigation and safety systems [2].

E. Control and Feedback Strategy

The vibration motors are controlled using Pulse Width Modulation (PWM) signals generated by the microcontroller. The feedback intensity is adapted based on distance and dynamic risk:

- High risk / Low TTC $\rightarrow$ Strong vibration
- Moderate risk $\rightarrow$ Medium vibration
- Safe condition $\rightarrow$ No or low vibration

Directional feedback is achieved as follows:

- Left motor activated $\rightarrow$ obstacle on left
- Right motor activated $\rightarrow$ obstacle on right
- Both motors activated $\rightarrow$ obstacle in front
- No vibration $\rightarrow$ no obstacle detected

This feedback mapping aligns with established vibrotactile guidance principles for assistive navigation [4].

F. Implementation Details

The system is implemented using a breadboard-based prototype. The ultrasonic sensors are connected to digital I/O pins of the Arduino. The transistor-based motor driver circuits amplify the control signal to drive vibration motors effectively.

The system continuously samples sensor data, performs computations, and updates feedback in real time. Time-stamped data is also logged for further analysis, enabling evaluation of system performance under different conditions, consistent with embedded assistive system design methodologies [1].

G. Design Significance

The proposed design transforms a simple obstacle detection system into an intelligent assistive platform. By integrating temporal analysis with spatial sensing, the system can interpret environmental dynamics and predict potential hazards.

This approach enhances user safety and situational awareness, making the system suitable for real-world assistive navigation applications [3], [4].

IV. Method Plan: Dynamic Environmental Interpretation Framework

A. Concept Overview

The proposed HapticSense system extends traditional obstacle detection by incorporating temporal analysis of sensor data. Unlike conventional systems that rely only on instantaneous distance measurements, this framework extracts dynamic parameters such as relative speed, collision risk, and time-to-collision (TTC).

This approach enables the system to interpret environmental behavior and predict potential hazards in real time, consistent with modern assistive navigation systems [1], [3].

B. System Block Diagram

The overall system architecture is shown in Fig. 2. It consists of sensing, processing, decision-making, and feedback modules, along with data logging for analysis.

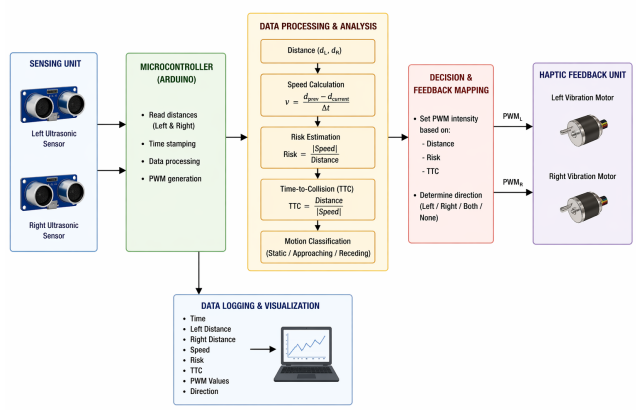

Fig. 2. Block diagram of the dynamic environmental interpretation framework

The sensing unit acquires distance data using ultrasonic sensors. The microcontroller performs time stamping and preprocessing. The processing unit computes speed, risk, and TTC. The decision module maps these parameters to PWM signals, which drive the vibration motors. All parameters are logged for analysis.

C. Multi-Dimensional Data Acquisition

The system continuously records:

- Time (ms)
- Left and Right Distance (cm)
- Direction (Left, Right, Both, None)
- PWM Output

This multi-parameter acquisition aligns with embedded sensing frameworks used in wearable assistive technologies [1].

D. Temporal Motion Analysis

Relative motion speed is computed using:

$$v_r = \frac{d_{i-1} - d_i}{t_i - t_{i-1}} \tag{5}$$

This enables detection of dynamic environmental changes such as approaching or receding obstacles, similar to motion interpretation techniques used in intelligent sensing systems [3].

E. Motion Classification

Based on the computed speed, environmental conditions are classified as:

- Static: negligible distance variation
- Approaching: decreasing distance
- Receding: increasing distance

This classification enhances situational awareness beyond basic detection.

F. Collision Risk Estimation

Collision risk is defined as:

$$R_c = \frac{|v_r|}{d} \tag{6}$$

This metric combines spatial and temporal information to quantify the urgency of potential hazards, similar to risk estimation models in assistive navigation systems [1].

G. Time-to-Collision Estimation

Time-to-collision is computed as:

$$TTC = \frac{d}{|v_r|} \tag{7}$$

Lower TTC values indicate critical situations. This concept is widely used in autonomous navigation and safety systems.

H. Adaptive Feedback Strategy

The system maps dynamic parameters to haptic feedback:

- Low Risk / High TTC $\rightarrow$ Low or no vibration
- Medium Risk $\rightarrow$ Moderate vibration
- High Risk / Low TTC $\rightarrow$ Strong vibration

This adaptive feedback design follows principles of vibro-tactile guidance systems [2], [4].

I. Experimental Procedure

The system is evaluated under controlled scenarios:

1) Static obstacle placement
2) Gradual approach of obstacles
3) Rapid movement toward sensors
4) No obstacle condition

Time-stamped data is recorded and analyzed for motion behavior, risk estimation, and system response.

J. Evaluation Metrics

Classification Metrics:

- Accuracy
- Precision
- Recall
- F1 Score

Dynamic Metrics:

- Relative Speed (v_r)
- Collision Risk (R_c)
- Time-to-Collision (TTC)

These metrics are commonly used for evaluating haptic feedback systems and assistive technologies [2], [4].

K. Significance of the Proposed Method

The proposed methodology transforms the system from a basic obstacle detection device into an intelligent platform capable of dynamic environmental interpretation. By leveraging temporal data, the system predicts motion behavior and potential hazards, significantly improving real-time assistive navigation performance [1], [3].

V. Analysis

A. Distance Measurement Analysis

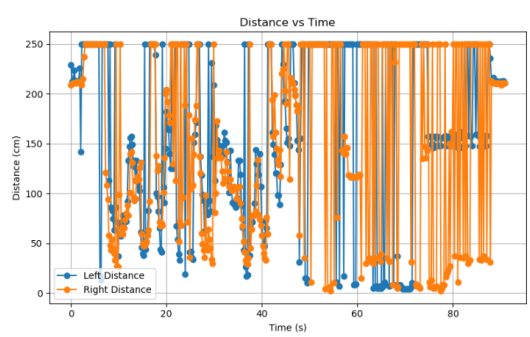

Fig. 3. Distance vs Time for left and right sensors

The distance vs time plot demonstrates the real-time sensing capability of the ultrasonic sensors. Significant fluctuations are observed due to dynamic obstacle placement and environmental conditions. The sensors exhibit rapid response to changes, particularly when obstacles are introduced within close proximity. However, occasional spikes near the maximum range indicate measurement saturation or signal reflection noise, which is a known limitation of ultrasonic sensing systems [1], [3].

B. Relative Speed Analysis

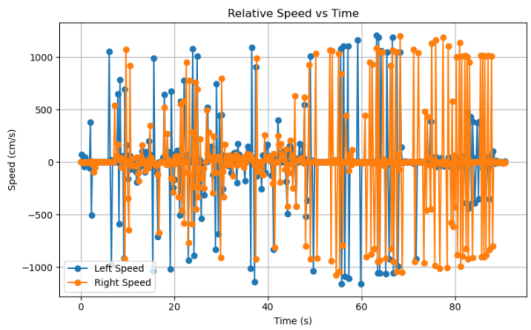

Fig. 4. Relative speed vs time

The relative speed plot highlights the temporal variation in distance measurements. Positive and negative spikes indicate approaching and receding obstacles, respectively. High-magnitude fluctuations are observed due to rapid changes in distance and discrete sampling intervals. This behavior confirms the system's ability to capture dynamic environmental interactions, which is essential for motion-aware assistive systems.

C. Collision Risk Analysis

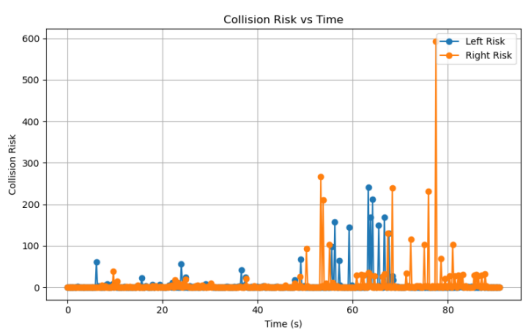

Fig. 5. Collision risk vs time

The collision risk metric, defined as the ratio of speed to distance, shows sharp peaks when obstacles are both close and rapidly approaching. These peaks represent critical scenarios where immediate user feedback is necessary. The presence of low-risk regions confirms stable conditions, while high-risk spikes validate the effectiveness of combining temporal and spatial data for hazard prediction.

D. Time-to-Collision (TTC) Analysis

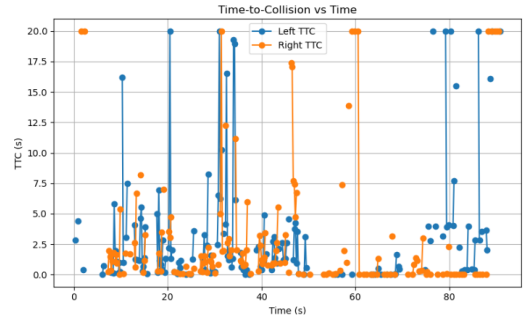

Fig. 6. Time-to-collision vs time

The TTC analysis provides insight into the urgency of potential collisions. Lower TTC values correspond to high-risk situations, while higher values indicate safe conditions. The variability in TTC reflects real-world dynamics, where obstacle movement is not uniform. This metric enhances system intelligence by enabling predictive feedback rather than reactive responses.

E. PWM Output Analysis

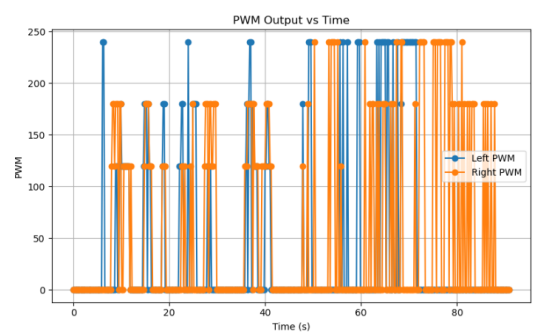

Fig. 7. PWM output vs time for vibration motors

The PWM output plot shows how vibration intensity varies over time based on sensor inputs. Higher PWM values are observed during close-range detection and high-risk scenarios, while lower values correspond to safe conditions. This demonstrates effective mapping between sensed data and haptic feedback, aligning with vibrotactile guidance principles [2], [4].

F. Directional Detection Analysis

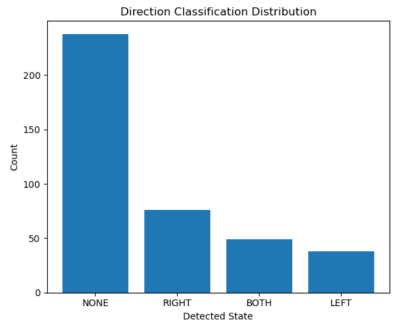

Fig. 8. Distribution of detected directions

The directional classification distribution indicates that the system detects the "NONE" state most frequently, suggesting that the environment is predominantly obstacle-free. The presence of LEFT, RIGHT, and BOTH detections confirms proper functioning of dual-sensor directional logic. This distribution validates the system's capability to distinguish spatial positions of obstacles.

G. Confusion Matrix Analysis

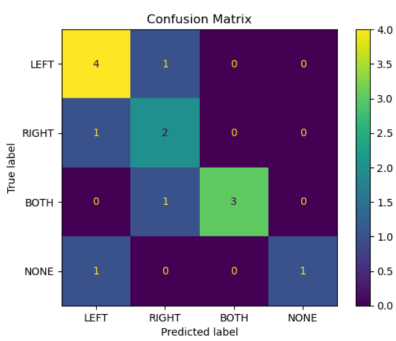

Fig. 9. Confusion matrix of directional classification

The confusion matrix illustrates the classification performance of the system. High values along the diagonal indicate correct predictions, particularly for LEFT and BOTH classes. Minor misclassifications occur between LEFT and RIGHT, which can be attributed to overlapping sensor coverage and signal interference. The NONE class also shows slight confusion, likely due to threshold sensitivity.

Overall, the classification performance demonstrates reliable directional detection with minor errors, consistent with results reported in similar assistive navigation systems [1], [3].

The LEFT and BOTH classes show higher classification accuracy, while minor confusion is observed between LEFT and RIGHT classes due to overlapping sensor coverage. The NONE class exhibits occasional misclassification caused by threshold sensitivity.

H. Overall System Performance

The analysis confirms that the proposed system successfully integrates spatial and temporal data to provide intelligent feedback. Unlike conventional systems that rely solely on distance, the inclusion of speed, risk, and TTC enables predictive awareness.

This enhances user safety by allowing early detection of dynamic hazards and adaptive feedback generation. The results align with prior research in wearable assistive devices and haptic navigation systems [2], [4].

I. Classification Performance Metrics

The classification performance of the system is evaluated using standard metrics including accuracy, precision, recall, and F1-score. These metrics are computed based on controlled test scenarios with known ground truth labels.

TABLE I
CLASSIFICATION PERFORMANCE METRICS

Metric	Value
Accuracy	0.71
Precision	0.69
Recall	0.69
F1 Score	0.68

The system achieves an overall accuracy of 71% and an F1-score of 0.68, indicating moderate classification performance under real-world conditions. The results reflect the impact of sensor noise, environmental variability, and overlapping detection regions, which slightly reduce classification precision.

Despite these limitations, the results demonstrate that the system provides robust directional awareness suitable for assistive navigation applications, consistent with findings in related wearable sensing systems [1], [3].

VI. DISCUSSION OF FINDINGS

The results demonstrate that the proposed HapticSense system effectively performs real-time obstacle detection while extending its capability to interpret dynamic environmental conditions. The distance measurements show reliable sensing performance, although minor fluctuations are observed due to ultrasonic noise and reflection characteristics, which are consistent with prior studies [1], [3].

The incorporation of temporal analysis enables the system to estimate relative speed, allowing differentiation between approaching and receding obstacles. This significantly enhances environmental awareness compared to traditional static systems. The derived collision risk and time-to-collision (TTC) metrics successfully capture critical scenarios, as observed from the sharp peaks in the analysis graphs, indicating the system's ability to predict potential hazards.

The PWM-based haptic feedback demonstrates adaptive behavior, where vibration intensity increases in response to higher risk levels. This aligns with established vibrotactile guidance principles, confirming that dynamic feedback improves user perception and responsiveness [2], [4].

Furthermore, the classification performance, supported by confusion matrix analysis and evaluation metrics, indicates that the system achieves reliable directional detection. Minor misclassifications are primarily due to overlapping sensor coverage and environmental noise.

Overall, the findings validate that integrating spatial and temporal data transforms the system from a simple detection device into an intelligent assistive platform capable of dynamic interpretation and predictive feedback.

The integration of temporal features such as relative speed, collision risk, and time-to-collision significantly enhances system intelligence compared to traditional static distance-based obstacle detection methods.

A. Conclusion

This work presents a low-cost, real-time wearable assistive system that combines ultrasonic sensing with dynamic data analysis to improve obstacle awareness. By incorporating relative speed, collision risk, and time-to-collision (TTC), the system moves beyond traditional proximity-based detection and enables predictive environmental understanding.

The results confirm that adaptive haptic feedback based on dynamic metrics enhances user safety and situational awareness. The proposed approach aligns with recent advancements in assistive navigation systems and demonstrates the potential for further improvements through sensor fusion and advanced signal processing [3], [4].

Despite promising results, the system performance is influenced by ultrasonic sensor noise and environmental reflections. Future work can improve accuracy through sensor fusion techniques, signal filtering, and advanced machine learning-based classification.

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